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STI Policy Review

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5~7th Fl, Building B, Sejong National Research Complex,
370 Sicheong-daero, Sejong, 30147, Republic of Korea

Tel: +82-44-287-2039
Fax: +82-44-287-2067
Web: http://www.stepi.re.kr

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Ⅰ Background and Rationale for the Research

Need to transition from a global agenda follower to a pioneering leader

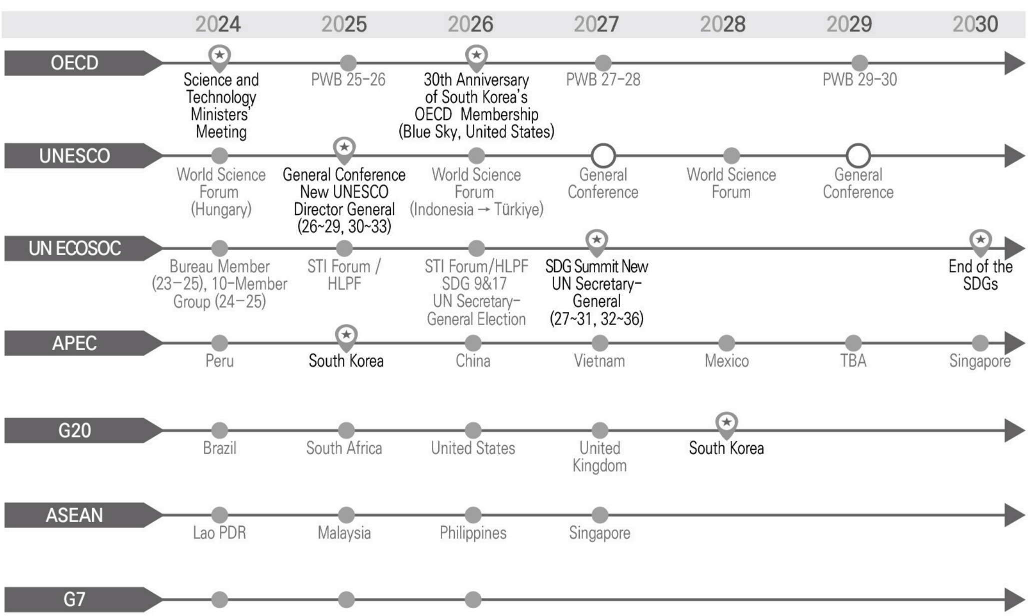
- Need to contribute to global agenda-setting and norm formation commensurate with the status of South Korea as an advanced science and technology nation: Although South Korea has grown into a science and technology (S&T) powerhouse, its role in global agenda-setting and norm formation remains limited.
  - By passively accepting and responding to agendas led by advanced nations, a disconnect has emerged between global agendas and domestic priorities. Consequently, South Korea faces a "global agenda follower's dilemma," in which its influence during the agenda-setting stage remains constrained.
  - This pattern of late participation also places South Korea at a disadvantage in the development of international technical standards and S&T norms, resulting in lagging agenda implementation and diminished international influence.
  - The Korean government has established, as a national policy objective, the goal of becoming a "strategic actor that leads global agendas" through pragmatic diplomacy that reflects national interests and strengthens international cooperation.

Need for a strategy to leverage policy windows in multilateral forums for post-2030 global S&T agenda-setting

- The Sustainable Development Goals (SDGs) are scheduled to conclude in 2030.
  - To achieve the 17 goals adopted by the international community between 2016 and 2030, United Nations (UN) Member States have emphasized SDG implementation through the active use of and cooperation in S&T, including emerging technologies.

- Growing attention to the visions of newly appointed heads of international organizations that will influence post-2030 global S&T agenda-setting
  - (UNESCO) Election of the first Arab Director-General in November 2025; 4-year term (2026-2029), with a maximum of 8 years (2026-2033) if reappointed.
  - (UN) Election of a new UN Secretary-General scheduled for 2026; term of office from 2027 to 2031.
- Opportunities for South Korea to propose agendas: 30th anniversary of membership in the Organization for Economic Co-operation and Development (OECD) in 2026, participation in the UN SDG Summit in 2027, and assumption of the G20 Presidency in 2028.

• **Figure 1** • Policy Windows in Major Multilateral Forums



Notes: PWB: Programme of Work and Budget, HLPF: High-Level Political Forum, UN ECOSOC: United Nations Economic and Social Council, APEC: Asia-Pacific Economic Cooperation, ASEAN: the Association of Southeast Asian Nations.

Source: Prepared by the research team.

Changes in the global science and technology landscape

- **Paradigm shift in science diplomacy:** The traditional view of scientific knowledge as a global public good has been weakening, while the tendency to regard S&T as a core national strategic asset has been strengthening. Consequently, cooperation and competition coexist, heightening the need for multifaceted S&T diplomacy strategies that simultaneously address national interests and global challenges.
- **Emergence of new alignments and governance issues:** As the cohesion of the Global South grows stronger, ethical concerns and governance issues surrounding emerging technologies have emerged as key agendas for the international community. These discussions are leading to the development of international technical standards and norms for emerging technologies.
- **Global leadership vacuum:** As competition between the United States and China and bloc formation among like-minded countries increasingly affects the landscape of multilateral S&T cooperation, the boundaries of cooperation are being redefined. Growing inward orientation among traditional agenda-shaping countries-driven by factors such as the United States' "America First" approach and worsening fiscal conditions in Europe-has reduced their capacity to engage with global issues, rendering a leadership vacuum in the international community increasingly apparent.

II Research Scope and Methodology

This study focuses on the global science, technology, and innovation (STI) agenda. A literature review is conducted, supplemented by the following research methodologies.

- Analysis of global STI agenda mechanisms and development of a global STI agenda-setting framework through action research
  - Collected and analyzed data by participating in government delegations to major multilateral forums and engagement in global discussions:
    - Collected and analyzed information on the latest global STI agendas; launched the "Global STI Agenda Hub," an information-sharing platform; and developed a global STI agenda-setting framework based on an analysis of global STI agenda mechanisms (who discusses what agendas, how, and where, and how these agendas evolve into international norms).
    - Contributed to discussions on the governance of emerging technologies and the development of international norms for research security by participating in the bureaus and expert committees of international organizations.
- Derivation of three post-2030 agenda clusters through horizon scanning
  - The OECD (2024) three-stage anticipatory governance framework (horizon scanning-structured preliminary assessment-extended assessment) is applied as a research methodology to establish a structured pathway from technology identification to policy decision support:
    - In the current year of the study (Year 2), the horizon scanning and structured preliminary assessment stages are being conducted as a pilot. In the next year, the study plans to deepen policy and governance alternatives through extended assessment (foresight, technology assessment, risk assessment, and scenario analysis).

- Horizon scanning is defined as a proactive tool for identifying and analyzing weak signals beyond the scope of current policy radar systems, enabling the early detection of technological and societal changes that may develop into future socioeconomic opportunities or threats.
- This study employs expert consultation, non-traditional data sources (e.g., patents, technology foresight, and keyword analysis), and large-scale web-based data and AI tools to broadly explore emerging technological and societal issues that are not captured by conventional statistics.
- During the structured preliminary assessment stage, strong signals (issues that are highly certain in the short term) are excluded, while strategic weak signals are identified and prioritized based on five criteria: intensity, familiarity, potential disruptiveness, potential scope, and likelihood of occurrence.
- Specifically, weak signals are defined with a focus on “high-impact, high-probability” issues, thereby moving beyond conventional conceptions of weak signals that only emphasize uncertainty and enabling the proactive concentration of policies and resources on strategic niche areas with a high likelihood of realization.
  - Scope of analysis: In total, 91 documents on STI agenda trends in multilateral forums and global foresight reports published between 2023 and 2025—including reports from the OECD, UN, European Union (EU), World Economic Forum (WEF), global consulting firms, futures research institutes, and major national governments—were analyzed, resulting in the identification of 383 future issues spanning all S&T and societal domains.
    - Through large-scale text summarization and classification using generative AI (ChatGPT), duplicate and similar issues were consolidated and standardized, reducing the 383 identified issues to 150 core issues. These were subsequently reclassified into three global STI agenda clusters based on technological domain, application area, and societal impact.
  - Derivation of three agenda clusters: AI and Digital Innovation; Environment, Energy, and Social Infrastructure; and Security Governance and Space

- The developed global agenda-setting framework is applied to conduct a pilot-stage preliminary identification of candidate agendas within the "AI and Digital Innovation" cluster.

Environmental analysis of three major post-2030 agenda clusters through expert consultation and interviews

- Established a global AI policy expert committee and high-level advisory committee:
  - Pilot-stage preliminary derivation of candidate agendas for "AI and Digital Innovation" through expert consultations.
  - Development of agenda-shaping strategies and recognition of the importance of proactive preparation for global STI agendas through consultations with high-level experts from government, academia, international organizations, and industry.

Ⅲ Key Research Content

Analysis of global science, technology, and innovation agenda mechanisms and latest trends

- Establishing the concept of global STI agenda-shaping:
  - Definition of global agenda-shaping: A set of actions through which a country elevates issues aligned with its national interests into shared international concerns and influences the formation of international norms and order through deliberations within institutionalized multilateral forums.
  - Analytical framework: Application of the concept of venue shopping to identify strategic behavioral patterns in agenda mechanisms by examining policy images and institutional venues as key components.
- Necessity of global agenda-shaping: Policy reinterpretation of the first-mover advantage (FMA) strategy
  - Application of the FMA strategy from technological innovation and corporate strategy perspectives to the global STI governance context.
  - Technological leadership can be interpreted as agenda-setting authority, preemption of assets as network effects, and buyer switching costs as leverage and lock-in effects. The advantages and disadvantages of a global agenda-shaper are as follows:

• **Table 1** • Advantages and Disadvantages of a Global Agenda-Shaper

Advantages	Disadvantages
Agenda-setting authority (preemption of normative frameworks and interpretive authority)	Free-rider issue (cost-free learning and adoption of improved models by latecomer countries)
Network effects (securing dominance through strategic alignment with key partners)	Technological policy uncertainty costs (risks associated with policy experimentation and trial-and-error implementation)
Leverage effects (enhanced bargaining power in diplomatic, economic, and technological negotiations through elevated national status)	Technological policy switching costs (structural transition costs arising from path dependency)
Lock-in effects (long-term structural dominance through norm institutionalization)	Normative backlash and political resistance (formation of opposing coalitions within the international community and declining trust)

Source: Prepared by the research team

- Proposed hybrid strategy: two-track approach of selective leader + active follower
  - Selective leader: Lead agenda-setting in areas of comparative advantage, introduce emerging agendas, and expand global leverage
  - Active follower: Maximize benefits within norms led by other actors and perform the role of a rule-shaper

• Latest trends in the global STI agenda

- OECD Committee for Scientific and Technological Policy (CSTP)
  - The STI Outlook 2025 emphasizes structural transformation and the reconfiguration of scientific cooperation amid geopolitical tensions (3P + 3 principles).
  - South Korea has participated in leading governance discussions on emerging strategic technologies by co-hosting Korea-OECD workshops on quantum technology and synthetic biology, and by voluntarily contributing funds to the OECD initiative on research security and research integrity.

- UNESCO Science Sector
  - In March 2025, the Ministerial Meeting on Science Diplomacy sought to further develop the concept of science diplomacy while balancing open science and research security.
  - In the first half of 2025, implementation reports were reviewed for three key recommendations (Recommendation on Science and Scientific Researchers [60 Member States], Recommendation on the Ethics of Artificial Intelligence [70 Member States], and Recommendation on Open Science [77 Member States]), while the Recommendation on the Ethics of Neurotechnology was newly adopted in the second half of the year.
  - The first Arab Director-General, Khaled El-Enany, was elected, presenting the vision of "UNESCO for the People" and placing humans at the center of the organization's mission. In particular, he emphasizes ethics, inclusiveness, and responsibility in digital transformation, AI, and emerging technologies, as well as the science-policy-society nexus and application of science in climate, environmental, and Earth sciences.
- UN 4th International Conference on Financing for Development (FfD4)
  - The "Seville Commitment" adopted in July 2025 elevated STI and digital technologies as key instruments for mobilizing development finance.
  - Agreement was reached on key areas including creating an STI policy environment, strengthening competition in digital markets, technology transfer and knowledge sharing, AI governance, and bridging the digital infrastructure gap.
- Asia-Pacific Economic Cooperation (APEC) Policy Partnership on Science, Technology and Innovation (PPSTI)
  - Established in 1990, it serves as a key mechanism coordinating STI cooperation in the Asia-Pacific region.
  - Under the 2016-2025 Strategic Plan, it promotes innovation-driven economic growth through collaboration among government, academia, and the private sector, focusing on three sub-groups: building science capacity, innovation-

enabling environments, and enhancing regional S&T connectivity.

- A new strategic plan is expected to be adopted from 2026 onward.

- **G20 Research and Innovation Working Group (RIWG)**
  - Officially launched under Brazil's 2024 presidency, the RIWG serves as a platform to address the innovation gap between the Global North and South.
  - South Africa's 2025 presidency selected "Equity in Science and Innovation-based Approach to Sustainable Development" as the theme.
- **ASEAN Plan of Action on Science, Technology and Innovation (APASTI) 2026-2035**
  - In June 2025, a new 10-year plan was adopted under the vision of an "integrated ASEAN powered by STI."
  - A mission-oriented approach was introduced, centered on five strategic objectives: implementation, sustainability, innovation, resilience, and inclusiveness, as well as future-oriented domains such as new technologies, energy transition, the blue economy, food systems, and space.
- **Others: BRICS**
  - Expanded membership (11 member states as of 2025 and 10 partner countries) has strengthened solidarity among the Global South, growing to a scale representing nearly half the world's population.
  - Cooperation is expanding into S&T fields beyond economic cooperation, such as health, digital technologies, AI innovation, and climate change response, which is expected to act as a new variable in South Korea's global S&T cooperation and governance engagement.

Post-2030 STI agenda derivation framework

- **(Candidate agenda clusters)** Derivation of post-2030 STI candidate agenda clusters
  - Horizon scanning from an anticipatory governance perspective yielded three clusters
    - **Cluster 1 ("AI and Digital Innovation")** encompasses issues related to expanded human-AI collaboration, superintelligence and human augmentation, and digital transformation issues such as the metaverse, digital twins, and quantum

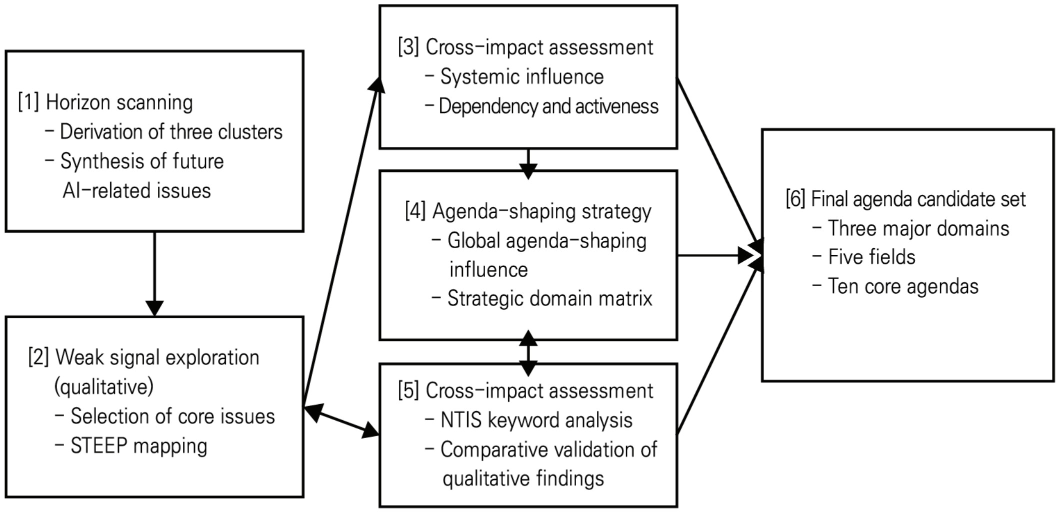
computing. It also entails socio-institutional challenges, including AI ethics, data governance, and labor market transitions.

- **Cluster 2 ("Environment, Energy, and Social Infrastructure")** covers technology and policy agendas aimed at addressing the climate crisis and strengthening societal infrastructure resilience, such as net-zero transitions and energy security, the circular economy, climate-resilient infrastructure, future food and water systems, and green finance.
- **Cluster 3 ("Security Governance and Space")** focuses on agendas at the intersection of advanced technologies and international security/global governance, such as the space economy and space security; emerging security threats based on AI, quantum, and biotechnology; and research security alongside international norms and ethical frameworks.

- Interdependence among the three clusters
  - The three clusters do not represent independent domains; instead, they form an interdependent large-scale system.
  - For example, AI and digital innovation serve as core solutions for climate and energy transitions, while the security governance cluster plays a supporting role in ensuring that such innovations secure social trust and sustainability.
- Accordingly, post-2030 global STI strategies should embed human-centered anticipatory governance—such as ethics, safety, and research security—from the initial stages of technological development and be designed and operated to maximize strategic synergies among the three clusters.

- **(Pilot-stage preliminary derivation of candidate agendas) Pilot-stage preliminary derivation of medium- to long-term global agendas within the "AI and Digital Innovation" cluster** using weak signal exploration based on horizon scanning
  - Stage 1 — Issue collection: 383 future issues were identified from 91 global foresight reports, which were reduced to 150 issues and classified into three clusters (AI/digital innovation, environment/energy, and security/space).

• **Figure 2** • Framework for Identifying Medium- to Long-Term Science, Technology, and Innovation Agendas Using Weak Signal Exploration Based on Horizon Scanning



Source: Prepared by the research team

- Stage 2 — Strategic weak signal exploration: 34 issues were derived from 100 AI-related reports, after which signal characteristics (novelty, evidential basis, maturity, urgency) and strategic significance (impact, policy relevance, disruptiveness, salience) were assessed through expert evaluation. Consequently, 17 core issues were selected, including 9 high-priority strategic targets and 8 trend transition targets.
- Stage 3 — Systemic influence assessment: Activeness and dependency are measured through a cross-impact analysis of the 17 issues and classified into 5 linkage issues (system core), 4 driving issues (strategic leverage), 4 dependent issues (monitoring targets), and 4 autonomous issues (independent response).
- Stage 4 — Assessment of global agenda-shaping potential: Based on systemic influence (Y-axis) and global agenda-shaping influence (X-axis: technology 0.5 + policy 0.3 + market 0.2), four strategic domains were derived high-differentiation drivers (5 items, e.g., AI semiconductors), moonshot missions (4 items, e.g., post-quantum cryptography security), hidden champions (4 items, e.g., smart grids), and smart followers (4 items, e.g., precision medicine).

- Stage 5 — Quantitative validation: National Science and Technology Information Service (NTIS) keyword analysis (2021-2025) was used to detect weak signals across 80 sub-topics. Alignment with qualitative analysis (multimodal AI), need for recalibration (AI drug discovery), and significant divergence (cybersecurity) were identified.
- Final output: A post-2030 AI-focused global STI agenda candidate set was proposed, consisting of three domains (AI-climate paradox, systemic disasters, redefinition of human roles), five fields, and ten core agendas.

• **Table 2** • Overview of Post-2030 AI Global Agenda Candidate Set

Three domains	Five fields	Ten agendas	Strategic alignment	Key content
A. AI climate paradox: Sustainable AI infrastructure and environmental governance	A.1 Computing infrastructure as a security asset	A.1.1 Semiconductor breakthrough strategy	Breakthrough driver	Expansion of next-generation memory and on-device AI chip investment to improve computational efficiency and reduce energy consumption, while simultaneously securing technological sovereignty and energy sovereignty
		A.1.2 Mandatory Green AI transition	Moonshot mission / Hidden champion	Legislation of responsible AI design to reduce energy consumption in data centers in parallel with smart grid deployment and energy security
	A.2 Leadership in the digitalization of environmental governance	A.2.1 Preemption of MRV standards	Smart follower	Leadership in AI-based carbon reduction MRV standard-setting and securing of credibility in green finance by establishing blockchain-based transparency platforms
		A.2.2 Preparedness for renewable energy integration	Hidden champion	AI-based renewable energy integration and automated water resource management governance advanced in coordination with higher-level issues, such as smart grids

Three domains	Five fields	Ten agendas	Strategic alignment	Key content
B. Systemic disaster risk nexus: Leading comprehensive AI security strategies and defensive technology	B.1 Preparedness for future security zero-day threats	B.1.1 PQC transition and CPS defense	Moonshot mission / Breakthrough driver	Establishment of a PQC transition roadmap in response to quantum computing threats and development of intelligent CPS security infrastructure
		B.1.2 Leadership in LAWS norms	Smart follower	Active participation in the formation of international norms on lethal autonomous weapons systems (LAWS) and clarification of the human-in-the-loop principle
C. Redefinition of human roles: Reforming the labor structure and establishing social inclusion systems	C.1 Redefine the meaning of future labor and knowledge production	C.1.1 Labor and social safety net redesign	Breakthrough driver / Hidden champion	Establishment of labor market policies and reskilling and upskilling systems in response to the proliferation of autonomous AI agents and humanoid robots
		C.1.2 Discussion of autonomous agent accountability	Breakthrough driver / Moonshot mission	Establishment of standards for the responsibility, ethics, and safety of AI agents, and introduction of validation platforms for autonomous AI research laboratories (XAI)
	C.2 Response to threats to human dignity and social trust systems	C.2.1 Reconstruction of trust in an AI-coexisting society	Hidden champion	Fact-checking support based on retrieval-augmented generation (RAG) to address AI hallucinations and deepfakes, and institutionalization of mandatory media literacy education
		C.2.2 Inclusive healthcare and care systems	Smart follower / Moonshot mission	Integration of AI-based precision medicine and care robots into welfare and healthcare policy, alongside the establishment of ethical guidelines and training of specialized personnel

Notes: MRV: Measurement, Reporting and Verification, CPS: Cyber-Physical Systems, PQC: Post-Quantum Cryptography

Source: Prepared by the research team

Cluster environmental analysis of the post-2030 global science, technology, and innovation agenda

- AI and digital innovation

(1) Generative AI and energy and environment

- Background and significance of emergence
  - AI climate paradox: The rapid advancement of generative AI induces substantial energy consumption and carbon footprints, with approximately 552 tons of CO_2 emissions reported for GPT-3 training (equivalent to driving a car 19 laps around the Earth). Furthermore, generative AI query processing consumes more than 10 times the electricity of a web search. Additionally, AI can potentially contribute to carbon reduction through improving energy system efficiency and climate prediction, thereby positioning the harmonization of digital innovation and green transition as a global agenda.
- Key issues for discussion
  - Improving energy efficiency of AI infrastructure: Reducing the electricity consumption of data centers and the use of cooling water, transitioning to clean energy, and encouraging the design of energy-efficient AI algorithms.
  - Environmental impact measurement and transparency: Most cloud companies do not disclose data on carbon emissions and water usage from AI services. Thus, developing reliable measurement indicators has become an urgent need.
  - Sustainable AI policies and norms: Incorporating environmental sustainability into AI strategies and discussing regulations such as mandatory disclosure of environmental impacts. The UNESCO Recommendation on the Ethics of Artificial Intelligence (2021) includes the principle of environmental responsibility, and the EU AI Act (2024) introduces provisions requiring mandatory recording of energy use and carbon emissions.
- Positions of major countries
  - EU: Establishes sustainability as a core principle of AI regulation, leading global standards by legislating energy efficiency and carbon emission transparency through the AI Act.

- United States: Directs the establishment of standards for clean energy use while maintaining a policy stance that encourages voluntary private-sector improvement. The 2025 AI Action Plan explicitly notes the relaxation of environmental regulations for the construction of data centers.
- China: Emphasizes technological development at the national level while externally highlighting the use of AI for sustainable development. The country also promotes cooperation with the Global South and green digital transformation.

- Future development directions
  - Global Digital Compact (2024) includes principles of environmental sustainability for digital technologies.
  - COP30 (November 2025) is expected to address decarbonization of the AI sector and the energy efficiency of data centers.
  - The OECD AI Compute and Climate expert group has been established to share best practices and develop policy recommendations.
  - Key future agenda items: ① carbon information disclosure for AI services, ② energy-efficient AI architectures, ③ renewable energy-based data centers, and ④ new research on environmental impacts such as AI water usage.

- Domestic perspective
  - Status: Data center infrastructure is rapidly expanding, but institutional mechanisms for the environmental impact of AI remain insufficient. The Korea Emissions Trading System (K-ETS) does not explicitly designate data centers or AI computation as separate regulated entities.
  - Challenges: Focus on AI infrastructure development versus insufficient emissions management in the digital sector under the carbon neutrality strategy. Need to align policy goals between “carbon neutrality by 2050” and “becoming a global top 3 AI semiconductor powerhouse by 2030.”
  - Opportunities: Potential to develop energy-efficient AI hardware based on a globally competitive semiconductor industry. A joint green AI agenda through

platforms such as P4G can be proposed based on experience as a bridge between advanced and developing countries.

(2) AI and data governance

- **Background and significance of emergence**
  - Rapid increase in strategic value of data: AI algorithms are based on an open ecosystem, but access to large volumes of high-quality data is a key determinant of competitiveness. While technologically advanced countries emphasize data openness and the free flow of data, latecomer countries are strengthening protectionism due to data sovereignty and security issues.
  - At the G20 Osaka Summit (2019), the leaders of the United States and China openly clashed over issues of data control, marking the emergence of a new major issue in international relations.
  - Data bias and privacy: When trained on biased data, AI systems can exacerbate social inequality through discrimination against specific racial or gender groups. Generative AI also poses risks of exposing personal information through unauthorized data collection, reuse, and hallucinations.
- **Key issues for discussion**
  - Bias mitigation and fairness: Ensuring the representativeness of training data, developing bias detection and correction technologies, and enhancing explainability of AI outputs. Cultural and legal systems differ across countries, leading to divergences in standards for defining bias and approaches to its remediation.
  - Privacy protection and secure data use: The large-scale unstructured data collection in generative AI is often conducted without the consent of data subjects, making existing protection frameworks such as the General Data Protection Regulation (GDPR) difficult to effectively operate. Efforts are underway to explore data mobility mechanisms that ensure privacy and security under the concept of "Data Free Flow with Trust" (DFFT, G7).

- Data sovereignty and international data mobility norms: Claims of national jurisdiction and control over data (China's Data Security Law (2021)) versus the guarantee of cross-border data flows (United States and EU). Key issues remain the establishment of legitimate jurisdictional boundaries, cross-border data access for law enforcement purposes, and the justification of data localization measures.

- Major multilateral forums and positions of major countries

- OECD: 2019 AI Principles (fairness, transparency, accountability) published. Operates the Global Partnership on Artificial Intelligence (GPAI) working group on data governance.
- UNESCO: The Recommendation on the Ethics of Artificial Intelligence (2021) includes data governance principles (non-discrimination, privacy protection).
- G7 Hiroshima AI Process (2023): Cooperation on generative AI governance, establishment of an international code of conduct in 2024, and launch of an OECD-led reporting framework (February 2025).
- G20: Emphasizes digital inclusion and public digital infrastructure under the Brazilian presidency (2024), reflecting the data sovereignty of the Global South.
- UN Global Digital Compact (2024): Includes principles on responsible data use and sharing, privacy protection, and AI ethics principles.

- Positions of major countries

- EU: Establishes global standards through GDPR and legislates mandatory data management obligations for high-risk AI systems under the AI Act (2024).
- United States: Prioritizes industry self-regulation and innovation. Released the AI Bill of Rights (2022) and the Executive Order on the Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence (2023).
- China: Strengthens political control through security reviews for cross-border data transfers and generative AI regulations (2023).
- Japan: Leads the DFFT initiative and cooperates with the EU to build a trusted data space.

- **Future development directions**
  - Global cooperation and normative competition proceed in parallel: The UN Global Digital Compact (2024) functions as a reference point for discussions.
  - G7: Promotes AI ethics and human rights as core agenda items under the Canadian presidency (2025) and issues the “G7 Leaders’ Statement on AI for Prosperity.”
  - G20: Emphasizes the reduction of the digital divide and data sovereignty of the Global South under Brazil (2024), forming a complementary discourse with DFFT.
  - OECD: Develops an AI risk management framework and voluntary reporting framework supporting implementation of the G7 Code of Conduct.
  - ISO/IEC SC42: Develops technical standards on explainability, bias mitigation, and privacy protection.
  - GPAI: Data definition and trust, public-interest data use, and development of trust certification tools.
- **Domestic perspective**
  - International participation: Adoption of OECD AI Recommendations, founding member of GPAI, EU GDPR adequacy decision (2021), and participation in the United States-EU Digital Partnership and the Bletchley Park AI Safety Summit.
  - Legal and institutional framework: Amendment of the Three Data Acts (2020, pseudonymized data utilization); AI Ethics Guidelines (2020); and Framework Act on AI (passed December 2024, implemented January 2026), the world’s first integrated legislation combining AI national strategy, promotion, and regulation, representing a Korean-style balanced model between the EU and the United States.
  - Infrastructure: Data Dam initiative, AI Hub, data standardization guidelines, quality certification system, and AI Safety Research Center (Established in November 2024).

- Capabilities: World-class ICT infrastructure, third globally in AI patents, and leading capabilities in semiconductors and telecommunications.

- **Implications**
  - Rule-maker potential: Leading implementation of UNESCO and OECD principles and proposing international technical guidelines to mitigate data bias.
  - Multi-layered middle-power diplomacy: A values-based data governance alliance grounded in freedom, human rights, and democracy (United States-EU-Japan), along with cooperation to reduce the digital divide and formation of an interest-based coalition with the Global South (developing countries) through capacity-building support leveraging ODA (Official Development Assistance).
  - Asian leadership: Leading regional data mobility principles within ASEAN+3 and APEC, leveraging the Korea-ASEAN digital innovation partnership.
  - International standardization: Active participation in ISO/IEC and sharing technologies for implementing open data and AI ethics as global public goods.
  - Institutional export: Sharing a balanced innovation-regulation model, including the Framework Act on AI, in international forums.

- **Environment, energy, and social infrastructure**
  - **International perspective**: Geopolitical transformation of climate issues
    - As climate change has emerged beyond an environmental challenge into a central agenda of geopolitical and industrial power competition, key multilateral forums such as the G7 and G20 have focused discussions on clean energy transition and securing critical mineral supply chains.
    - The G7 has set targets to transition to 100% zero-emission new vehicles by 2035, triple renewable energy capacity, and secure 1,500 GW of energy storage capacity by 2030. The UN and OECD only emphasize transition principles to protect workers in fossil fuel industries.
  - **Emerging issues**: 23 emerging issues across 6 domains

- Clean energy transition: Six issues, including green hydrogen, small modular reactors (SMRs), smart grids, next-generation batteries, and electric vehicles. The G7 has proposed international standards for low-carbon hydrogen and set a 1,500 GW target for batteries, while the OECD is promoting international cooperation on SMR safety standards.
- Industrial decarbonization: Five issues, including sustainable data centers; securing the Carbon Capture, Utilization, and Storage (CCUS) of critical minerals; synthetic fuels; and the circular economy. The G7 has established a transparent critical mineral supply chain and adopted common Circular Economy and Resource Efficiency Principles (CEREP), while the OECD has developed policy recommendations for green ICT.
- Agriculture, food, and water: Five issues including alternative proteins, precision agriculture, vertical farming, the WEF nexus (Water-Energy-Food nexus), and AWH (Atmospheric Water Harvesting). The UN is discussing the contribution of agricultural technology innovation to food security, and the G20 is promoting cooperation to reduce the digital agriculture divide.
- Ocean ecosystems: Two issues, including the blue economy and blue carbon. The G7 reaffirms implementation of the "Ocean Deal" and the goal of protecting 30% of the ocean by 2030, while the UN promotes technological cooperation through the "Decade of Ocean Science."
- Governance of emerging technologies: Three issues, including solar radiation modification (SRM), AI-based climate modeling, and bio-digital convergence. The UN has highlighted "AI and climate change," while SRM requires governance discussions due to concerns over regional climate disruption.
- System transition: Two issues, including green economic transition and carbon markets. The G7 has reaffirmed its commitment to circular economic transformation for net zero by 2050, while the OECD recommends inclusive carbon market development measures.

- Roles of major multilateral forums

- OECD (designer of climate and industrial policy): The "Green Industrial Policy"

(2024) defines green subsidies as an essential tool for overcoming market failures such as infrastructure gaps and path dependency. The report, "What is Unique About Green Innovation?" (2025), analyzes five key characteristics of green technologies that distinguish them from conventional innovation, including green hydrogen and CCUS-linked green steel. The "Climate Action Monitor 2025" warns of slowing climate action across countries and insufficiency of current Nationally Determined Contributions (NDCs) in achieving the targets set in the Paris Agreement.

- UN (coordinator of global governance): The "Emissions Gap Report" of the United Nations Environment Programme (UNEP) indicates that a 57% reduction is required by 2035, but current NDCs would lead to a 2.3-2.5°C temperature increase by the end of the century, making a temporary overshoot of the 1.5°C target unavoidable. The "Adaptation Gap Report" notes that adaptation costs in developing countries are 5-10 times higher than actual financing, diagnosing a widening gap and providing critical scientific input for COP30 negotiations on the New Collective Quantified Goal on Climate Finance (NCQG).
- G20 (catalyst for bridging the North-South divide): Under the Brazilian presidency in 2024, climate was linked to poverty and inequality, and discussions on innovative financing mechanisms such as "wealth taxes" were formally institutionalized. At COP30 (Brazil, 2025), expected outcomes included raising the ambitions of NDCs toward 2035, specifying the NCQG at more than USD 100 billion annually, and the formal emergence of trade-related issues (the EU's Carbon Border Adjustment Mechanism [CBAM] and the United States' Inflation Reduction Act [IRA]).
- UNESCO (social and ethical foundation): "Education and Climate Change" (2025) calls for a paradigm shift in climate change education (CCE) from knowledge transmission to socially, emotionally, and action-oriented learning. It emphasizes the integration of local and indigenous knowledge and inclusive co-creation involving youth, indigenous peoples, and vulnerable groups, and calls for making climate education mandatory in all national curricula by 2025.

- **Four key issues of the international community**
  - [Issue 1] Trade/security: Competition over green subsidies and weaponization of supply chains: A global competition in green subsidies, triggered by the United States' IRA and the EU's Green Deal, is intensifying into a struggle for leadership in future energy technologies such as green hydrogen, next-generation batteries, and SMRs. China's control of approximately 80% of the battery manufacturing supply chain, along with tightened export controls on rare earths, graphite, and gallium, sustains the risk of supply chain weaponization.
  - [Issue 2] Finance/equity: Failure to meet the 1.5°C target and a dual gap: UNEP's formalization of the inevitability of a temporary overshoot of the 1.5°C target has triggered a credibility crisis in global governance. While advanced economies are channeling subsidies into domestic industries, developing countries face financing shortfalls amounting to 5-10 times their adaptation costs, exacerbating a dual gap. The COP30 NCQG negotiations are therefore emerging as the top priority for restoring trust.
  - [Issue 3] Technology/ethics: Governance gaps and adverse effects of frontier technologies: As the failure to meet the 1.5°C target increases interest in radical technologies such as solar radiation modification (SRM), concerns are rising over potential regional climate disruptions and the risk of international conflict, while governance frameworks remain absent. Unilateral deployment by technologically leading countries may result in disproportionate harm to technologically lagging countries, emerging as a new driver of North-South tensions.
  - [Issue 4] Society/transition: Operationalization of just transition and social pressure: As climate response is increasingly recognized as a whole-of-society transformation, just transition has rapidly emerged as a core agenda item for the G20, UN, and OECD. UNESCO emphasizes shifting CCE toward social and emotional learning and highlights the need to enhance societal acceptance of emerging technologies such as alternative proteins and smart farming.

- **Domestic policy trends**
  - Core elements of domestic policy: Based on the Framework Act on Carbon Neutrality and Green Growth for Coping with Climate Crisis (2021), South Korea has set targets of carbon neutrality by 2050 and a 40% emissions reduction by 2030. The 5th Energy Technology Development Plan (2024-2033) focuses on building a carbon-free energy (CFE) ecosystem including SMRs, clean hydrogen, next-generation batteries, and CCUS, while explicitly addressing energy supply chain vulnerabilities and the weaponization of critical minerals as part of an economic security approach.
  - Linkages with international issues: [Issue 1] Strong linkage (technology self-reliance in SMRs, hydrogen, and batteries, and carbon market institutionalization directly addressing green subsidy competition) / [Issue 2] Lack of linkage (limited positioning in COP30 NCQG due to the absence of discussions on adaptation finance in developing countries and the loss-and-damage fund) / [Issue 3] Neglect of linkage (lack of public discourse on governance and ethical gaps for frontier technologies such as SRM) / [Issue 4] Weakening of the social dimension (just transition limited to industrial workers, and limited implementation of CCE)
- **Tailored global agenda-setting strategy**
  - [Strategy A] OECD-centered “technology standards coalition”: Leading the establishment of transparent and fair technology standards such as SMR safety standards and clean hydrogen certification systems, and disseminating policy experience in carbon neutrality legislation and emissions trading systems
  - [Strategy B] UN-centered “adaptation technology assistance”: Transitioning to solution-based aid by packaging precision agriculture, smart farming, the WEF nexus, and AWH, and proposing ICT-based climate data sharing and early warning systems at the UN STI Forum
  - [Strategy C] G20-centered “frontier technology governance”: Establishing a position “opposing unilateral deployment” through anticipatory research on the potential impacts and ethical security risks of frontier technologies, and

leading international agreement on “principles of responsible research and transparency” within the G20

- [Strategy D] UNESCO-centered “digital CCE and just transition”: Modeling South Korea’s experience of “technology-society conflicts” in rapid industrialization into a policy framework, and proposing “digital climate education content” and “youth climate leader exchange programs” using ICT and EdTech through the UNESCO platform

- **Science and technology security**
  - **Background and significance of emergence**
    - The securitization of S&T has been driven by the competition between the United States and China for technological hegemony, triggering economic security issues and intensifying global technological competition, with S&T increasingly recognized as a strategic national asset.
    - As the arena of interstate technological competition expands to university campuses and basic research environments, major countries are actively institutionalizing risk management of S&T within an economic security framework.
    - Cases are increasing in which international research collaboration by general researchers becomes a threat to national and economic security. The exfiltration of key national research assets is occurring, alongside a growing risk of inappropriate research interference by foreign institutions that may threaten researchers.
    - Researchers are exposed to security risks without being aware of changes in geopolitical conditions and in the policy directions of governments and institutional partners abroad.
    - This raises growing concerns that it damages the openness of the global research ecosystem and the value of international cooperation, both of which are essential for S&T development and innovation promotion.

- **Key issues**
  - In the routine international cooperation of researchers and research institutions, is the free sharing of public R&D information and outputs appropriate?
  - The recent strengthening of S&T securitization conflicts with the core pillars of research and innovation promoted for decades, including international cooperation, research openness, and academic freedom. How can an appropriate balance be achieved?
  - Emerging strategic technology fields such as AI, quantum, space, and synthetic biology are areas in which most advanced S&T countries are concentrating substantial R&D investment for the future, while also experiencing intensifying inter-state competition. How should countries design strategies that balance promotion and protection in these fields, and how should they develop partnership approaches accordingly?
  - Can recent research security risks be mitigated through existing export controls, various economic sanctions within global value chains, and traditional classified research management systems in defense and security research? How do they differ, and what additional measures are required?
- **International community and positions of major countries**
  - When governments invest taxpayer-funded resources in R&D, they also assume the risks of potential failure. Therefore, it is considered unjust for certain countries to appropriate research outputs supported over many years through national R&D programs solely for their own national interests.
  - To safeguard the fundamental values of the research ecosystem and ensure fair and constructive competition, it is emphasized that each country should strengthen awareness of research security risks within the research community and take appropriate response measures in the event of security incidents.
  - **(Common response directions)** Although research ecosystems differ across countries and levels of research security risk also vary, the following common response directions are identified: (1) strengthening research integrity and

fostering a safe research culture based on trust, (2) systematic and continuous risk management of national R&D programs, and (3) management of country-specific vulnerabilities.

- **(OECD)** Systematization of the 3P (Promotion, Protection, Projection) approach and policy implementation monitoring: The OECD systematizes member states' policy implementation monitoring through the 3P approach. It published the 2022 policy report "Integrity and Security in the Global Research Ecosystem," established the OECD STIP Compass research security portal (updating member country trends every two years), and launched the 2025 Research Security and Research Integrity project (covering the concept and scope of research security, risk management frameworks, and spillover effects). It also addresses the securitization of S&T in the OECD Science, Technology and Innovation Outlook 2025.
- **(EU)** Application of the "as open as possible, as closed as necessary" principle: The EU ensures research openness and academic freedom while implementing research security measures when necessary, emphasizing "responsible international research." It issued the 2022 Marseille Declaration on International Cooperation in Research and Innovation and the 2024 Council Recommendation on enhancing research security and applies the EU Economic Security Strategy to Horizon Europe. It is also advancing the establishment of a European Centre for Research Security (2025) and exploring legally binding mechanisms for research security (ERA Act). The EU takes an integrated approach to research security, linking it with the internationalization of public research funding, research integrity, open research data, and ethics in emerging technologies.
- **(G7)** Since the Working Group on the Security and Integrity of the Research Ecosystem (SIGRE) was established in 2021 under the joint presidency of the United Kingdom and Canada, annual principles and model frameworks for research security have been derived, with efforts underway to extend them beyond G7 members toward global norms.

Table 3 Key G7 Research Security Activities

Year	Presidency	Key Content
2021	United Kingdom	Agenda-setting on research security and establishment of the SIGRE Working Group
2022	Germany	7 core values of research integrity and 8 principles of research security
2023	Japan	Research security risk management framework
2024	Italy	Best practice casebook on research integrity and research security
2025	Canada	Ongoing development of a standard model for research security in international collaborative research programs
2026	France	Expected establishment of a research security standard model

Source: Prepared by the research team

- **(United States)** "Research Security" is a core policy orientation for maintaining global leadership. The United States has adopted policies based on a sense of urgency to sustain STI global leadership and the need to prevent research security risks arising from strategic competitors. The country has advanced follow-up measures on research security based on a legislative and executive framework, including President Trump's Presidential National Security Memorandum-33 (NSPM-33, January 2021), amendments to the National Defense Authorization Act (NDAA), and the CHIPS and Science Act (2022). It has diversified its whole-of-government response (export controls, research integrity, international cooperation, and regulatory standardization) and established a virtuous-cycle stakeholder engagement mechanism through the National Academies of Science, Engineering, and Medicine (NASEM) Roundtable (outreach-consultation-policy improvement). The National Science Foundation (NSF) has established a research security division and operates the SECURE research security program (education and training, data collection and analysis). It is also advancing efforts to address regional disparities in research funding distribution and to increase STEM education and research funding in strategic industries (e.g., AI, quantum computing, advanced manufacturing, 6G, energy, materials science, etc.).
- **(Canada)** Ensuring a strong "Research Security" policy, the Sensitive

Technology Research & Affiliations of Concern (STRAC) policy, and effectiveness thereof: Canada introduced this policy against the backdrop of heightened awareness of (research) security risks across society and the economy, including the Nortel case re-investigation, intergovernmental tensions related to Huawei, and incidents involving the leakage of biological research samples from national research institutes. The STRAC policy applies to research projects in sensitive technology areas funded by Canada's three main research funding agencies (Canadian Institutes of Health Research [CIHR], Social Sciences and Humanities Research Council [SSHRC], and Natural Sciences and Engineering Research Council of Canada [NSERC]) and the Canada Foundation for Innovation. It requires researchers to submit a "sensitive technology research attestation" and imposes measures such as suspension or recovery of funding and rejection of future funding applications in cases of non-compliance. Eleven sensitive technology research areas (e.g., digital infrastructure, energy, manufacturing materials, weapons, AI, sensors, biotechnology, quantum, robotics, etc.) have been designated, and a list of 110 research partners of concern is maintained (89 Chinese institutions, with the remainder from Russia and Iran). Furthermore, Innovation, Science and Economic Development Canada (ISED) and Public Safety Canada (PSC) operate regional "research security centers," establishing clear guidelines and advisory and communication mechanisms to ensure effectiveness. A tailored approach by target group is being implemented (government: security, research institutions: safeguarding, researchers: "protect your research").

- **(United Kingdom)** "Trusted Research" is a key policy concept that focuses on establishing a trust-based research culture rather than strong regulatory control. University autonomy and institutional reputation are emphasized. The United Kingdom ranks 4th globally in international co-authored research output, with 62% of the country's research outputs resulting from international collaboration. China is the United Kingdom's second-largest international research partner, and Chinese students represent the largest group of international students in the country's universities, underscoring the importance of international research collaboration and talent mobility for national S&T competitiveness. The United Kingdom began its response following a 2019 parliamentary self-initiated inquiry, which identified inadequate management of research collaboration with China by government departments and universities. The central government strengthened existing related functions (Cabinet Office restrictions on overseas investment in 17 sensitive technology areas, Home Office restrictions on technology transfer in dual-use technology fields, Foreign Office strengthening of Academic Technology Approval Scheme (ATAS) review processes, and Education Department monitoring of university researchers' overseas income-generating activities). The Department for Science, Innovation and Technology established five regional Research Collaboration Advice Teams (RCATs). The National Protective Security Authority (NPSA) developed and disseminated tailored guidelines and checklists, and UK Research and Innovation (UKRI) specified three principles of trusted research and conducted compliance reviews of recipient institutions (piloted in the health and medical sector). The limited role of government is defined as (1) providing "trusted research" content, (2) offering advisory services and helpdesks (RCATs), and (3) focusing on key stakeholders.

- **Future development directions**
  - Norms and order formation in the global research ecosystem: The international community approaches research security not merely as a security issue, but also as a means to ensure the integrity, transparency, and openness of the entire research ecosystem. Accordingly, each country is implementing policies using context-specific keywords (research security in the United States and Canada, trusted research in the United Kingdom, foreign interference in Australia, research integrity in Japan, knowledge security in the Netherlands, and responsible internationalization in Sweden).
  - In the future, major multilateral forums such as the OECD, EU, and G7 are expected to advance the development of standardized models for research security risk management, the sharing of best practices, interoperable policy

frameworks, and linkages with and impact assessments of other policy areas (e.g., open science and governance of emerging strategic technologies). These efforts are expected to shape norms and order in the global research ecosystem.

- **Domestic status and implications**
  - **(Status)** In 2023, the Presidential Advisory Council on Science & Technology approved the "Measures for Enhancing the Soundness of the National Research System," followed by amendments to the National Research and Development Innovation Act, which introduced provisions requiring researchers to report foreign funding. As of 2025, continuous institutional improvements under the National Research and Development Innovation Act include the establishment of a new "sensitive project" classification requiring strategic investment management. Following the designation of sensitive countries by the United States Department of Energy, dedicated units for research security issues were established within the Ministry of Science and ICT (MSIT) and the National Research Foundation of Korea (NRF).
  - **(Issue)** As a national strategic asset, S&T is a cross-ministerial issue; however, the government-level response remains highly fragmented. In particular, since the enactment of the Act on Prevention of Divulgence and Protection of Industrial Technology in 2006, a need exists at this stage to integrate and harmonize the established, systematized industrial technology protection regime and stakeholder roles, and the newly emerging roles and responsibilities within the S&T community. South Korea's industrial technology protection system aims to safeguard national industrial technologies and maintain competitiveness in global technological competition. It is based on legal frameworks such as the Act on Special Measures for Strengthening the Competitiveness of, and Protecting, National High-Tech Strategic Industries; Foreign Trade Act; and Unfair Competition Prevention and Trade Secret Protection Act. Key actors include firms, intelligence agencies, investigative authorities, and law enforcement agencies, with responses primarily based on institutional and regulatory framework improvements and ex post facto

enforcement against violations. In contrast, research security aims to foster a trust-based and safe research ecosystem. It is grounded in legal and regulatory frameworks such as the National Research and Development Innovation Act, Security Measures for National R&D Programs, and Joint Management Regulations for National R&D Programs. Key actors include universities, public research institutes, and specialized research management agencies, with a policy focus on strengthening researchers' compliance with research integrity standards and managing research security risks in national R&D programs as a government directive. The approach emphasizes prevention through raising awareness and cultivating a safe research culture.

- **(Proposal)** S&T identifies "researchers and research assets" as objects of protection and requires the establishment of an integrated research security system encompassing international collaboration (personnel and academic exchange, and joint research), industry-academia collaboration (technology transfer, commercialization, and joint research), research ethics, and security (physical and technical security). At the global realignment level, it should contribute to science diplomacy reflecting national interests and to the formation of norms and order in the global research ecosystem.
- **(Implications for leading the global agenda)** The trend toward the securitization of S&T is expected to continue for the foreseeable future, requiring deeper consideration of Korea's role in shaping a new international order and norms in a global research ecosystem where cooperation and competition coexist. The United States-China rivalry and bloc formation among like-minded countries are also affecting multilateral S&T cooperation frameworks, leading to a redefinition of the boundaries of collaboration. Furthermore, increasing inward orientation among traditional agenda-setting actors, driven by United States prioritization and fiscal constraints in Europe, is reducing their capacity to engage in global issues, and a leadership vacuum has already begun to emerge. This presents both a challenge and an opportunity.

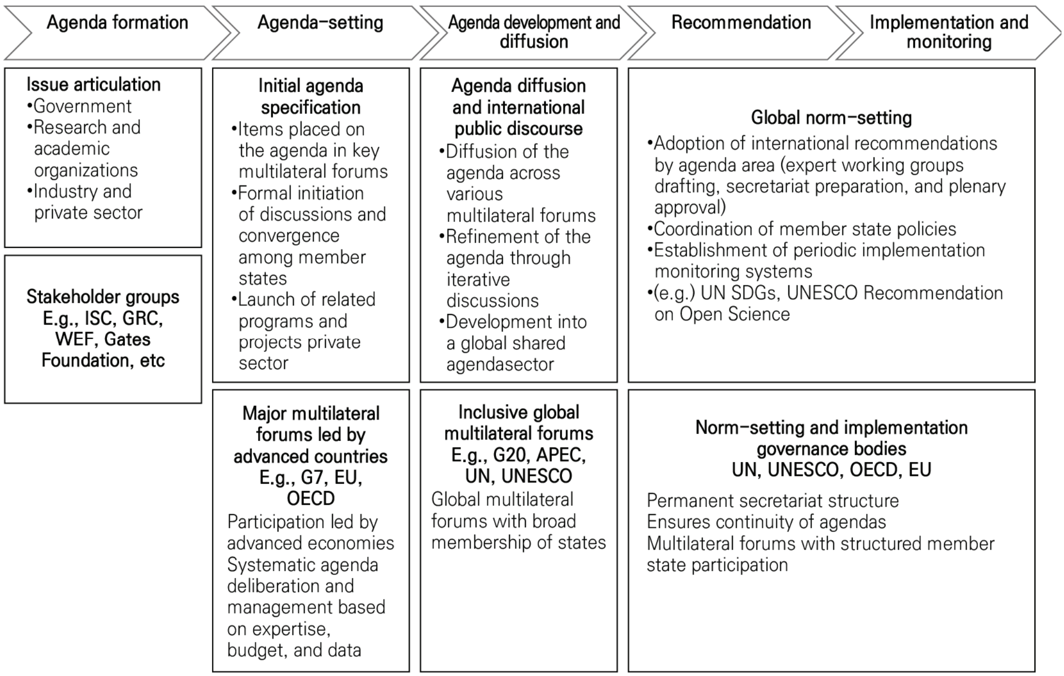
Ⅳ Conclusion and Policy Recommendations

Deriving tailored agendas for multilateral forums based on a global science, technology, and innovation agenda mechanism

- **Multilateral forum selection and linkage: Sequential and parallel agenda-shaping approaches**
  - Selecting a strategy linking forum (venue shopping): A strategy that goes beyond simple venue selection to linking venues; for example, Japan's Data Free Flow with Trust (DFFT) was successfully institutionalized through sequential linkage from the World Economic Forum to the Osaka G20 and Hiroshima G7 Leaders' Declaration.
  - Providing tailored agenda content and parallel approaches that consider the characteristics of multilateral forums.
    - OECD: Serves as a knowledge hub. It excels in proactive agenda-setting within the global agenda mechanism and maintains a system that covers the full cycle from agenda development, recommendation, and follow-up to implementation.
    - UNESCO: Has strengths in agenda dissemination and the global normative institutionalization of issues through the world's largest membership base and network of field offices.
    - G7: Agenda formation led by major advanced economies. Although not a member, engagement through G7+ provides opportunities for cooperation among like-minded countries at the early stage of agenda formation.
    - G20: South Korea can play the role of a middle power, a model country, and a coordinator. South Korea's hosting of the 2028 G20 represents a strategic opportunity.

- Furthermore, bilateral negotiations with major S&T countries should be linked to ensure consistent agenda-setting.

• **Figure 3** • Global Science, Technology, and Innovation Agenda Mechanisms



Notes: ISC: International Science Council, GRC: Global Research Council, WEF: World Economic Forum
Source: Prepared by the research team

- Strategic importance of agenda-shaping
  - Clear vision and purpose of agenda-shaping: In an era in which S&T has become a central issue in international politics and the global economy, global agenda-setting therein is a core element of national competitiveness and diplomatic leadership.
  - Selection of an intelligence-based domestic agenda: Agendas should originate from domestic capabilities and experience, and then be scaled up to the global level. Issues must be identified, understood, and diffused domestically so that Korea can sustain continuous contribution and engagement even after being

placed on the international agenda.

- The importance of naming and framing that provides directionally meaningful value at the intersection of national interests and global consensus is highlighted.

- **Tasks for strengthening South Korea's capabilities**
  - Domestic intelligence capacity: For effective agenda identification, content must already be identified, understood, and diffused domestically. A system is needed that enables the generation of ideas across diverse fields.
  - Excessive emphasis on issue-driven responses has resulted in a system that is strong in signaling domestic demand but weak in absorptive capacity. Strengthening the receiver function of think tanks and academia is required.
  - Furthermore, it is necessary to enhance the capacity of research communities to participate in core networks that lead the formation of a global agenda (e.g., Korean Academy of Science and Technology, Korean Federation of Science and Technology Societies, academic societies, and policy experts).
  - Linkage between technological sovereignty and agenda-shaping: Without securing technological capability, agenda-shaping lacks sufficient leverage.

A global science, technology, and innovation agenda-setting framework

- **Limitations of this study**
  - There are methodological limitations in deriving S&T agendas that reflect national interests while simultaneously projecting a long-term future society at the global level.
  - In the follow-up study, wherein agenda-setting work will be further developed, qualitative supplementation is planned through expert interviews and the operation of consultation meetings involving domestic and international experts.

Figure 4 Post-2030 Global Science, Technology, and Innovation Agenda-Setting (Derivation 〉 Strategy 〉 Implementation) Framework

Identifying candidat global agenda clusters	01 **Horizon scanning** ▸ Major multilateral agenda documents and global foresight reports
Identifying candidate global agenda list	02 **Weak signal exploration (qualitative)** ▸ Selection of key issues and mapping to STEEP (Social, Technological, Economic, Environmental, and Political) dimensions 03 **Cross-impact assessment** ▸ Systemic influence, importance, and activeness 04 **Agenda-shaping strategy** ▸ Matrix of global leadership influence and strategic domains 05 **Weak signal exploration (quantitative)** ▸ NTIS keyword analysis and comparison and supplementation of qualitative results
Deriving a global agenda-setting strategy	06 **Agenda ecosystem analysis** ▸ Mapping of key actors (academia, government, industry leadership), agenda development stages, and venue shopping (multilateral forums) 06 **Deriving a global agenda-setting strategy** ▸ Developing strategies by agenda type and tailored agenda packages for each multilateral forum
Implementing global agenda-setting	07 **Post-global agenda-setting institutionalization, development, and support for norm implementation** ▸ Proposal of agenda packages by forum, and promotion of multilateral cooperation using power, financial, intelligence, and diplomatic instruments

Source: Prepared by the research team

Authors

- Project Leader: Inkyoung Sun, Research Fellow, isun@stepi.re.kr
- Participants: Dongwoo Lee, Associate Research Fellow, dw459@stepi.re.kr
  Eunhye Hwang, Senior Administrator, hye0226@stepi.re.kr
  Eunjoo Kim, Principal Researcher, ejkim@stepi.re.kr
  Jihyun Kim, Research Fellow, jihyunkim@stepi.re.kr
  Jehyun Yu, Researcher, yu11@stepi.re.kr
  Aram Lee, Researcher, aramlee@stepi.re.kr
  Myunghwa Lee, Senior Research Fellow, mlee@stepi.re.kr
  Hwanil Park, Senior Research Fellow, hipark@stepi.re.kr
  Wangdong Kim, Senior Research Fellow, wangdkim@stepi.re.kr
  (External author) Jisun Hwang, Han-Sol Lee

SCIENCE AND TECHNOLOGY POLICY INSTITUTE

5~7th Fl, Building B, Sejong National Research Complex, 370 Sicheong-daero, Sejong, 30147, Republic of Korea
www.stepi.re.kr